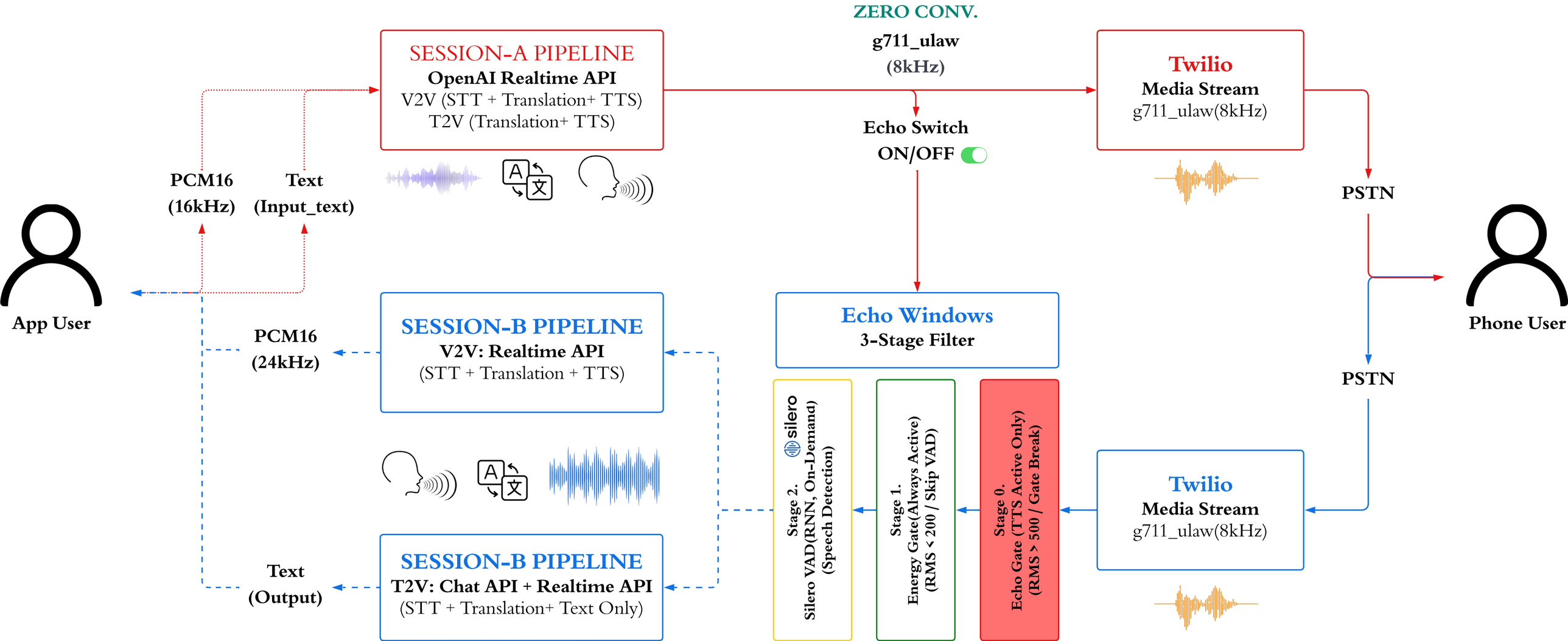
ZERO CONV.
g711_ulaw
(8kHz)
SESSION-A PIPELINE
OpenAI Realtime API
V2V (STT + Translation+ TTS)
T2V (Translation+ TTS)
Twilio
Media Stream
g711_ulaw(8kHz)
Echo Switch
ON/OFF
PCM16
(16kHz)
Text
(Input_text)
PSTN
App User
Phone User
PCM16
(24kHz)
SESSION-B PIPELINE
V2V: Realtime API
(STT + Translation + TTS)
Echo Windows
3-Stage Filter
PSTN
Stage 2.
silero
Silero VAD(RNN, On-Demand)
(Speech Detection)
Stage 1.
Energy Gate(Always Active)
(RMS < 200 / Skip VAD)
Stage 0.
Echo Gate (TTS Active Only)
(RMS > 500 / Gate Break)
Twilio
Media Stream
g711_ulaw(8kHz)
Text
(Output)
SESSION-B PIPELINE
T2V: Chat API + Realtime API
(STT + Translation+ Text Only)